



WAREHOUSE DELAY & DAMAGE ANALYTICS

End-to-End Supply Chain & Logistics Analytics Project

Project Type

Industry-Level Supply Chain & Logistics Analytics Project

Tools & Technologies

Python | AWS S3 | AWS RDS (MySQL) | SQL | Power BI | Power Query | DAX

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1. Executive Summary

Warehouse operations sit at the center of every e-commerce and logistics business. When shipments are delayed, damaged, or fail to meet promised delivery timelines, the impact is felt directly in operating costs, SLA penalties, and customer trust.

This project builds an end-to-end analytics solution — spanning Python, cloud infrastructure (AWS S3 and AWS RDS), SQL, and Power BI — to monitor warehouse operations, uncover the root causes of delays and damage, and track SLA compliance across warehouses, regions, delivery partners, and operating shifts.

Across roughly 25,000 shipment records, the analysis found a 63.51% delivery success rate, a 22.73% delay rate, a 24% damage rate, and SLA compliance of just 61.02% — with Delhi Warehouse and the North region consistently underperforming, and adverse weather emerging as the single largest driver of failed deliveries. The findings are delivered through three interactive Power BI dashboards designed to support fast, data-driven operational decisions.

2. Project Overview

This is an industry-level, end-to-end analytics project modeled on the operational challenges faced by large-scale delivery and logistics businesses. Rather than starting from an already-clean dataset, the project was built as a full pipeline: a public delivery-logistics dataset was enhanced with warehouse-specific business columns, cleaned and feature-engineered in Python, staged in AWS S3, loaded into a cloud-hosted MySQL database on AWS RDS, analyzed with SQL, and finally modeled and visualized in Power BI.

The result is a warehouse operations analytics solution that identifies bottlenecks, quantifies financial losses from damaged shipments, benchmarks warehouses and delivery partners against each other, and gives management a single, interactive view of delivery performance.

3. Business Problem

Modern logistics and e-commerce companies move enormous volumes of shipments through multiple warehouses, delivery partners, and regions every day. Despite this scale, several recurring operational problems are difficult to diagnose without structured analysis:

- Frequent shipment delays across warehouses and regions
- Damaged products during transportation and handling
- SLA (Service Level Agreement) breaches
- High and inconsistent operational costs
- Inefficient or underperforming warehouses

- Inconsistent delivery partner performance

Business teams need an analytics solution that provides real-time operational visibility and helps pinpoint exactly where — and why — these issues are occurring, so corrective action can be targeted rather than generic.

4. Business Objectives

- Monitor warehouse performance across delivery success, ratings, and failures
- Track shipment delays by warehouse, region, and delivery partner
- Analyze damaged shipments and quantify their financial impact
- Measure SLA compliance across the delivery network
- Identify operational bottlenecks caused by weather, shift, and vehicle type
- Improve logistics efficiency and reduce operational losses
- Support data-driven decision-making through interactive dashboards

5. Target Industry Use Cases

The project is designed around business scenarios similar to large-scale delivery and quick-commerce operations, including companies such as:

- Amazon
- Flipkart
- Blinkit
- Zomato
- Swiggy
- Delhivery
- Blue Dart
- DHL

These businesses share a common need: visibility into warehouse-level and partner-level delivery performance at scale.

6. Dataset Information

Source: Kaggle — Delivery Logistics Dataset (India, Multi-Partner)

The original dataset contains 25,000+ delivery operation records. It was enhanced with additional warehouse-specific business columns and a feature-engineered SLA status field to support warehouse-level operational analysis.

Additional Business Columns Added

- warehouse_id
- warehouse_name
- shift
- damage_status
- delay_reason
- damage_cost

Feature Engineered (via Python)

- sla_status — derived by comparing delivery_time_hours against expected_time_hours

Core Dataset Fields

- delivery_id, delivery_partner, package_type, vehicle_type, delivery_mode
- region, weather_condition, distance_km, package_weight_kg
- delivery_time_hours, expected_time_hours, delayed, delivery_status
- delivery_rating, delivery_cost, warehouse_id, warehouse_name, shift
- damage_status, delay_reason, damage_cost, sla_status

7. Tools & Technologies

Category	Tool / Technology
Programming	Python (Pandas, NumPy)
Database	MySQL
Cloud Storage	AWS S3
Cloud Database	AWS RDS (MySQL)
Query Language	SQL
Visualization	Power BI

Category	Tool / Technology
Data Transformation	Power Query
Business Measures	DAX
Data Modeling	Star Schema
Version Control	GitHub

8. Project Workflow & Architecture

The project follows a structured, 13-phase Business Intelligence / data engineering workflow, taking the data from a raw public dataset all the way to a documented, portfolio-ready analytics product:

- Phase 1: Business Understanding
- Phase 2: Dataset Analysis
- Phase 3: Python Data Cleaning
- Phase 4: Feature Engineering
- Phase 5: AWS S3 Upload
- Phase 6: AWS RDS Database
- Phase 7: SQL Analysis
- Phase 8: Power BI Data Modeling
- Phase 9: DAX Measures
- Phase 10: Dashboard Development
- Phase 11: Automation
- Phase 12: Documentation
- Phase 13: GitHub Deployment

Automation Flow

Raw Dataset → Python Cleaning Script → Feature Engineering → Processed Dataset → AWS S3 Upload → AWS RDS Update → Power BI Refresh

Automation is currently handled via Windows Task Scheduler, with AWS Lambda identified as a future enhancement for cloud-native scheduling.

9. Data Cleaning & Feature Engineering (Python)

Python (Pandas and NumPy) was used to load, inspect, clean, and enhance the raw dataset before it was staged for the cloud pipeline.

Key Python Tasks

- Data loading — reading the raw CSV into a Pandas DataFrame
- Initial exploration — shape, head/tail, column names, and datatypes
- Missing value analysis and duplicate detection
- Data type correction and text standardization
- Feature engineering of the sla_status column
- Export of the cleaned, feature-engineered dataset for upload to AWS S3

Feature Engineering — SLA Status

The existing sla_status column was dropped and rebuilt from first principles to ensure it accurately reflected delivery performance:

Logic: If delivery_time_hours > expected_time_hours → "Breached SLA", Else → "Within SLA"

10. Cloud Infrastructure — AWS S3 & AWS RDS

AWS S3 (Simple Storage Service)

AWS S3 was used as the cloud object storage layer for both the raw and processed datasets.

- Bucket: warehouse-delay-damage-analytics
- Region: Asia Pacific (Mumbai) — chosen for lower latency and better performance
- Bucket type: General Purpose (standard for CSV / analytics workloads)
- Object Ownership: ACL Disabled (bucket owner owns all objects — AWS-recommended)
- Block Public Access: Enabled, for data security
- Versioning: Disabled (not required for a portfolio project)

AWS RDS (Relational Database Service)

AWS RDS provided a fully managed, cloud-hosted MySQL database so the analytics data could be queried directly with SQL without maintaining a server manually.

- Engine: MySQL 8.4.8
- Configuration: Full (Standard) Configuration
- Template: Free Tier

- Deployment: Single-AZ (cost-efficient for a portfolio project)
- DB Instance Class: db.t4g.micro
- Storage: 20 GB General Purpose SSD

Data Flow

Python (cleaning & feature engineering) → CSV → AWS S3 → AWS RDS (MySQL) → MySQL Workbench → SQL Analysis → Power BI Dashboard

11. SQL Business Analysis

With the cleaned dataset loaded into AWS RDS, SQL was used to transform raw operational records into business-ready answers before any dashboard was built. Queries relied on CTEs, window functions (DENSE_RANK), aggregate functions, CASE-based bucketing, and percentage calculations to compare performance across warehouses, regions, delivery partners, shifts, and operational conditions.

The analysis was organized into nine themes: delay, damage, SLA, operational factors, cost, warehouse performance, delivery partner performance, customer experience, and failure / root cause analysis — covering 41 distinct business questions in total.

12. Business Questions Solved Using SQL

1. Delay Analysis (5 queries)

- Which warehouse has the maximum delays?
- Which region has the highest delays?
- Which delivery partner has the highest delayed deliveries?
- Delay percentage by warehouse
- Average delivery delay (hours)

2. Damage Analysis (5 queries)

- Overall damage percentage
- Total damage cost
- High-risk warehouses by damage rate
- Shift-wise damage counts
- Region-wise damage rate

3. SLA Analysis (5 queries)

- SLA compliance %

- SLA breach %
- Warehouse SLA performance
- Delivery partner SLA performance
- Region-wise SLA compliance

4. Operational Analysis (6 queries)

- Weather impact on deliveries
- Vehicle performance
- Average delivery time by delivery mode
- Average delivery time by region
- Shift-wise delivery performance
- Average delivery time by package type

5. Cost Analysis (5 queries)

- Average delivery cost by region
- Average delivery cost by vehicle type
- Cost vs. distance
- Cost vs. package weight
- Average delivery cost across delivery modes

6. Warehouse Performance Analysis (5 queries)

- Total deliveries by warehouse
- Warehouse rating comparison
- Warehouse with the highest failed deliveries
- Top-performing warehouse
- Warehouse delivery success rate

7. Delivery Partner Analysis (5 queries)

- Total deliveries by partner
- Average rating by partner
- Failed deliveries by partner
- Damaged deliveries by partner
- On-time delivery percentage by partner

8. Customer Experience Analysis (3 queries)

- Average delivery rating

- Rating by region
- Rating by delivery partner

9. Failure & Root Cause Analysis (3 queries)

- Delivery status distribution
- Reasons behind unsuccessful deliveries
- Weather vs. failed deliveries

13. Data Modeling — Star Schema

The cleaned data was modeled in Power BI using a star schema, separating the transactional fact table from descriptive dimension tables to keep the model fast and the DAX measures simple.

Fact Table

- Fact_Delivery

Dimension Tables

- Dim_Warehouse
- Dim_Region
- Dim_Vehicle
- Dim_Package
- Dim_DeliveryPartner
- Dim_Date

All relationships are one-to-many from each dimension table into the fact table.

Key DAX Measures

- Total Shipments
- Delayed Shipments
- Delay %
- Damage %
- Total Damage Cost
- Average Delivery Time
- SLA Compliance %
- Average Delivery Cost

14. Power BI Dashboard Development

Three interactive dashboard pages were built to give management a full view of operational health — from a high-level executive summary down to root-cause diagnostics and cost drivers. All pages share consistent Region, Warehouse, Delivery Partner, and Package Type slicers, and are connected through in-report navigation buttons.

Dashboard Flow: Executive Overview → Operational Insights → Cost & Performance

Dashboard 1 — Executive Overview

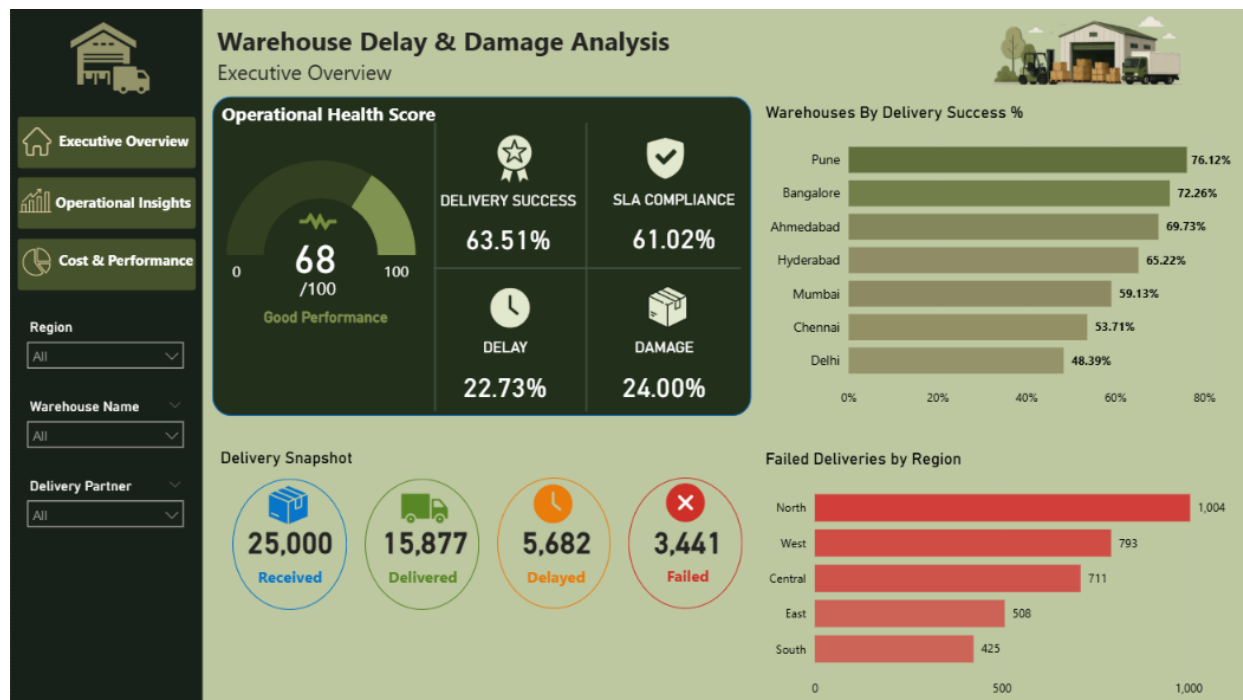


Figure 1: Executive Overview — Warehouse Delay & Damage Analysis

Purpose

Gives leadership a single-glance operational health score for the entire warehouse network, along with a snapshot of shipment volumes and regional failure hotspots.

Dashboard Includes

- Operational Health Score (0–100 gauge, currently 68/100 — "Good Performance")
- Delivery Success %, SLA Compliance %, Delay %, and Damage % scorecards
- Delivery Snapshot — Received, Delivered, Delayed, and Failed shipment counts
- Warehouses by Delivery Success % (bar chart, all 7 warehouses ranked)
- Failed Deliveries by Region (bar chart)

Business Value

Gives management the ability to assess overall network health in seconds: of 25,000 shipments received, 15,877 were delivered, 5,682 delayed, and 3,441 failed, with the North region contributing the most failures (1,004).

Dashboard 2 — Operational Insights

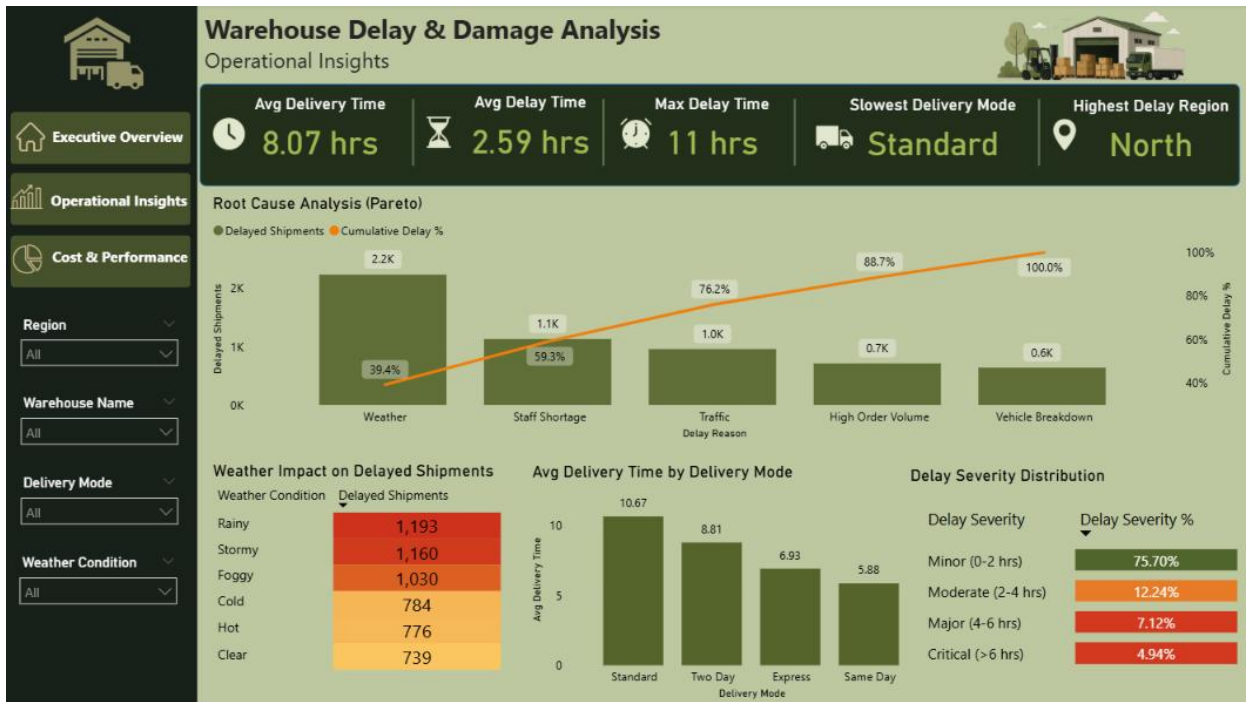


Figure 2: Operational Insights — Warehouse Delay & Damage Analysis

Purpose

Drills into why shipments are delayed, using a Pareto (root-cause) view alongside weather and delivery-mode breakdowns.

Dashboard Includes

- Avg Delivery Time, Avg Delay Time, Max Delay Time scorecards
- Slowest Delivery Mode and Highest Delay Region callouts
- Root Cause Analysis (Pareto chart) — delayed shipments by cause with cumulative delay %
- Weather Impact on Delayed Shipments (table by weather condition)
- Average Delivery Time by Delivery Mode
- Delay Severity Distribution (Minor / Moderate / Major / Critical)

Business Value

Identifies Weather as the single largest root cause (2.2K delayed shipments, ~39% of the total, 76% cumulative with Staff Shortage), and shows Standard delivery mode as the slowest at 10.67 hrs average — pinpointing exactly where operational investment will have the most impact.

Dashboard 3 — Cost & Performance

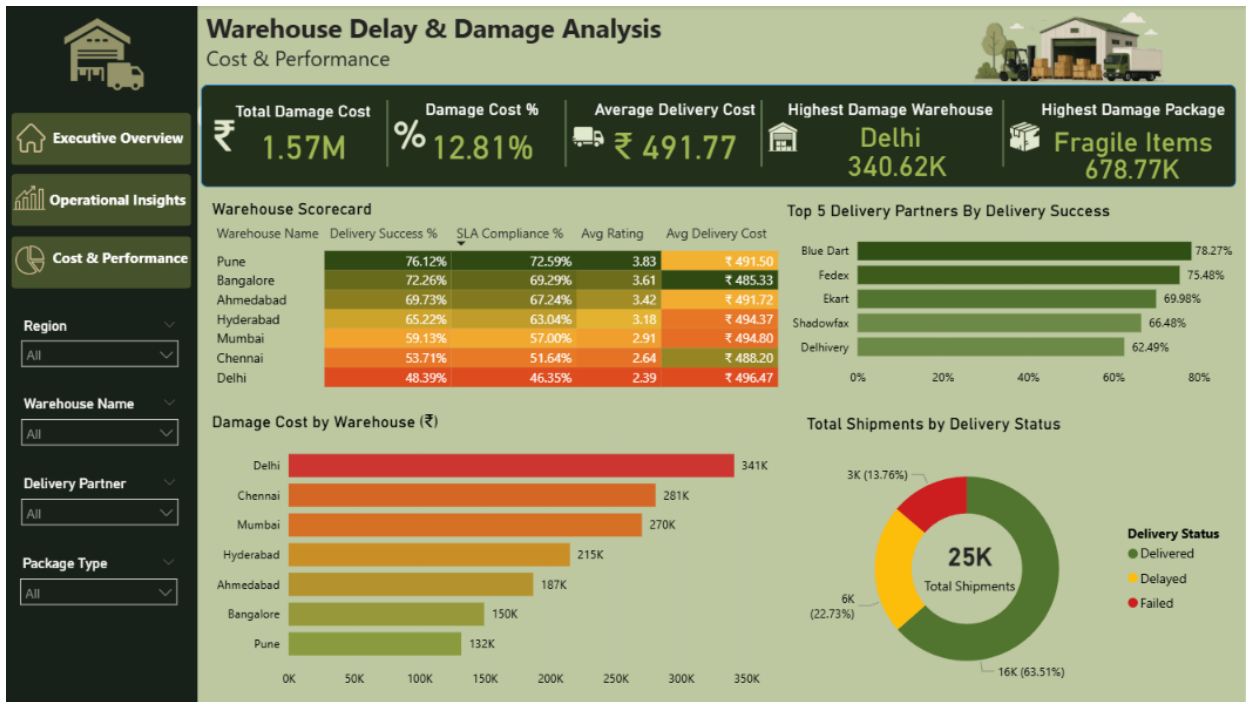


Figure 3: Cost & Performance — Warehouse Delay & Damage Analysis

Purpose

Quantifies the financial side of returns and damage, and benchmarks warehouses and delivery partners against each other.

Dashboard Includes

- Total Damage Cost, Damage Cost %, Average Delivery Cost scorecards
- Highest Damage Warehouse and Highest Damage Package Type callouts
- Warehouse Scorecard — Delivery Success %, SLA Compliance %, Avg Rating, Avg Delivery Cost per warehouse (conditionally formatted)
- Top 5 Delivery Partners by Delivery Success %
- Damage Cost by Warehouse (bar chart)
- Total Shipments by Delivery Status (donut chart)

Business Value

Surfaces that Delhi Warehouse drives the highest damage cost (₹340.62K) and the weakest scorecard (48.39% delivery success, 46.35% SLA compliance), while Pune leads on every metric — giving management a clear best-practice benchmark and a clear intervention target.

15. Dashboard Features

Interactive Slicers

Users can filter every page by Region, Warehouse Name, Delivery Partner, Package Type, and Weather Condition.

Navigation Buttons

In-report navigation lets users move seamlessly between Executive Overview, Operational Insights, and Cost & Performance.

Dynamic KPIs

All KPI cards recalculate automatically based on the selected filters, giving real-time operational insight.

Conditional Formatting

Color-coded formatting (red / amber / green) highlights high-risk warehouses, delivery partners, and regions at a glance — visible in the Warehouse Scorecard and Weather Impact tables.

Root Cause (Pareto) Analysis

A Pareto chart combines delay-cause volume with cumulative percentage, immediately showing which few causes account for most delays — a classic 80/20 diagnostic view.

16. KPI Summary

KPI	Business Purpose
Total Shipments	Measures overall delivery demand
Delivery Success %	Measures overall delivery quality
Delay %	Measures the share of shipments delivered late
Damage %	Measures the share of shipments damaged in transit
Total Damage Cost	Measures financial loss caused by damaged shipments
SLA Compliance %	Measures how often promised delivery timelines are met
Average Delivery Time	Measures typical delivery speed
Average Delivery Cost	Measures typical per-shipment logistics cost
Operational Health Score	Composite score summarizing overall network performance

17. Key Business Insights

Delay Insights

- Chennai Warehouse recorded the highest number of delayed deliveries (891), with a warehouse-wise delay rate of 24.95% — the highest in the network.
- The North region recorded 1,391 delayed deliveries, the most of any region.
- Ecom Express recorded the highest delayed deliveries among delivery partners (813 shipments).
- The average delay for delayed shipments is approximately 2.59 hours.

Damage Insights

- The overall shipment damage rate is 24%, with total damage cost reaching ₹1,574,965.
- Delhi Warehouse has the highest damage rate at 37.10%, followed by Chennai (31.95%) and Mumbai (28.16%).
- Night shift recorded 2,978 damaged shipments — far above Morning (1,546) and Evening (1,476), pointing to reduced supervision or handling issues after hours.
- The North region has the highest regional damage rate (27.68%); the South region the lowest (20.43%).

SLA Insights

- Overall SLA compliance is 61.02%, meaning nearly 39% of shipments breach their promised delivery window.
- Pune Warehouse leads SLA compliance at 72.59%; Delhi Warehouse trails at 46.35%.
- Blue Dart leads delivery partners on SLA compliance (74.66%); Ecom Express trails (49.12%).
- The South region leads regional SLA compliance (70.10%); the North region trails (50.05%).

Operational Insights

- Rainy (1,193) and Stormy (1,160) weather conditions drive the most delayed deliveries.
- Standard delivery mode has the longest average delivery time (10.67 hrs) versus Same Day (5.88 hrs).
- The North region has the highest average delivery time (8.59 hrs); the South region the lowest (7.56 hrs).
- Night shift deliveries average 8.58 hrs versus 7.63 hrs for Morning shift.

Cost Insights

- The North region has the highest average delivery cost (₹530.79); the South region the lowest (₹444.85).
- Truck deliveries cost the most on average (₹619.02); bike deliveries the least (₹406.26).
- Delivery cost rises consistently with distance and package weight — Very Long Distance shipments average ₹677.05 versus ₹303.84 for Short Distance.
- Same Day delivery is the most expensive mode on average (₹605.20) versus Standard (₹414.09).

Warehouse & Delivery Partner Insights

- Pune Warehouse leads on delivered volume (2,719), average rating (3.95), and delivery success rate (76.12%).
- Delhi Warehouse has the most failed deliveries (1,021) and the lowest success rate (48.39%).
- Blue Dart leads delivery partners on volume (2,190 delivered), rating (3.76), and on-time delivery (78.27%).
- Xpressbees has the most failures (664) and damaged shipments (914) among partners, with the lowest rating (~3.07).

Customer Experience & Root Cause Insights

- The overall average delivery rating is 3.47; the South region rates highest (3.73), the North lowest (3.07).
- Delivered shipments account for 63.51% of the total; delayed 22.73%; failed 13.76%.
- Weather is the leading root cause behind unsuccessful deliveries (3,909 occurrences), ahead of Staff Shortage (1,769), Traffic (1,461), High Order Volume (1,058), and Vehicle Breakdown (926).
- Stormy weather alone accounts for 1,610 failed deliveries, versus just 111 for Clear weather.

18. Business Recommendations

Warehouse Improvement

- Give Delhi Warehouse immediate operational intervention — a dedicated SLA and damage-reduction plan, plus a packaging and handling audit.
- Review Chennai Warehouse for processing bottlenecks, dispatch delays, and workforce allocation.
- Document Pune Warehouse's operating practices and use them as the network benchmark.

Delivery Partner Management

- Prioritize Blue Dart for high-priority, time-sensitive shipments.
- Put Xpressbees and Ecom Express on a structured improvement plan with measurable SLA, damage, and rating targets.
- Track all partners on scorecards covering on-time %, failure rate, damage rate, SLA compliance, and rating.

Operational Improvements

- Integrate weather forecasting into delivery planning, with backup capacity for rainy and stormy conditions.
- Strengthen Night shift supervision and quality checks to reduce damage.
- Add preventive vehicle maintenance to reduce breakdown-related failures.
- Improve workforce planning to reduce staff-shortage-driven delays.

Business Strategy

- Prioritize North region operations — it underperforms on delay, damage, cost, SLA, and customer rating simultaneously.
- Reflect Same Day delivery's higher operational cost in its pricing.
- Monitor SLA, delay, damage, failure, cost, and rating KPIs continuously via the Power BI dashboards rather than periodic manual reporting.

19. Business Impact

- Full visibility into refund-equivalent losses from damaged shipments
- Clear identification of high-risk warehouses, partners, and regions
- Root-cause-level understanding of delivery failures, not just symptom-level KPIs
- Faster operational reporting through interactive, self-service dashboards
- A repeatable, cloud-based data pipeline (S3 → RDS → Power BI) rather than a one-off spreadsheet analysis

- A performance benchmark (Pune / Blue Dart) that other warehouses and partners can be measured against

20. Skills Demonstrated

Technical Skills

- Python (Pandas, NumPy) for data cleaning & feature engineering
- AWS S3 for cloud object storage
- AWS RDS (MySQL) for cloud-hosted relational databases
- SQL — CTEs, window functions, aggregate functions, CASE-based bucketing
- Power BI, Power Query, and DAX
- Star schema data modeling

Analytical & BI Skills

- Root cause / Pareto analysis
- Warehouse and delivery partner benchmarking
- SLA and operational KPI development
- Cost driver analysis
- Dashboard design and data storytelling
- End-to-end analytics pipeline design (cloud data engineering + BI)

21. Project Outcomes

- Built a complete cloud-based data pipeline: Python → AWS S3 → AWS RDS (MySQL) → SQL → Power BI
- Answered 41 distinct business questions across 9 analytical themes using SQL
- Delivered 3 interactive Power BI dashboards covering executive overview, operational root-cause analysis, and cost & performance benchmarking
- Identified Delhi Warehouse, the North region, and Night shift operations as the top intervention priorities
- Quantified ₹1.57M in total damage cost and a 38.98% SLA breach rate as key financial and service risks

22. Future Enhancements

- Automate the pipeline with AWS Lambda instead of Windows Task Scheduler
- Add a dedicated Delivery Partner Scorecard dashboard page
- Introduce predictive delay/damage risk scoring using machine learning
- Add real-time data refresh from AWS RDS into Power BI Service
- Extend the model with a profitability / cost-to-serve dashboard
- Add anomaly detection alerts for sudden spikes in delay or damage rate

23. Conclusion

The Warehouse Delay & Damage Analytics project demonstrates a complete, cloud-native Business Intelligence workflow — from raw data ingestion through Python cleaning, AWS S3 and RDS infrastructure, SQL-based business analysis, and finally interactive Power BI dashboards.

The analysis converts 25,000+ raw delivery records into concrete, prioritized action items: Delhi Warehouse and the North region need immediate operational attention, Night shift handling needs closer supervision, and weather-aware delivery planning could meaningfully reduce the 22.73% delay rate and 13.76% failure rate seen in the data.

Beyond the specific findings, the project reflects the kind of end-to-end pipeline — data engineering plus analytics plus visualization — that real logistics and e-commerce organizations rely on to run their operations.

24. Project Deliverables

- Raw and cleaned/processed datasets
- Python data cleaning & feature engineering script
- AWS S3 bucket setup and configuration notes

- AWS RDS (MySQL) database setup and configuration notes
- SQL scripts — table creation and 41 business-analysis queries
- Power BI data model (star schema) and DAX measures
- Power BI dashboard file (.pbix) — 3 interactive pages
- Dashboard screenshots
- Project documentation (this document)

25. Live Dashboard

The interactive Power BI dashboard can be accessed using the link below.

Live Dashboard:

<https://app.fabric.microsoft.com/view?r=eyJrIjoieMjBjMjkwZDMtMGQ0Ni00Nzc1LTkyNmItZDlwYTlmZGVkOTI5IiwidCI6IjllZGM5ZjZmLTc4YjUtNDZjMS1iYzkwLTNkZGM0NjExNWE4NiJ9>

The dashboard includes interactive filters for Region, Warehouse Name, Delivery Partner, and Package Type, allowing users to explore warehouse performance, delay and damage root causes, and cost trends dynamically.